SOMAIYA
VIDYAVIHAR
K J Somaiya Institute of Technology
An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Course Name and Code: Data Structures Lab **(**ITL302)
Semester: III (SYIT)
Academic Year: 2026-27 (Odd Semester)
Name: Rashi
Roll No: 23

Experiment No. 04

Aim: To implement a Double-Ended Queue (Deque) using an array for a real-world Heatwave Monitoring System, allowing temperature readings to be inserted and deleted from both ends.

Objectives:
1.To understand the concept of a Double-Ended Queue (Deque).
2.To implement a Deque using an array.
3.To perform insertion and deletion from both ends of the queue.
4.To accept temperature readings as user input.
5.To apply the Deque concept to a real-world heatwave monitoring system.

Theory:A Double-Ended Queue (Deque) is a linear data structure in which insertion and deletion operations can be performed from both the front and rear ends.

In this experiment, an array is used to store temperature readings collected by a heatwave monitoring system.

The main operations are:

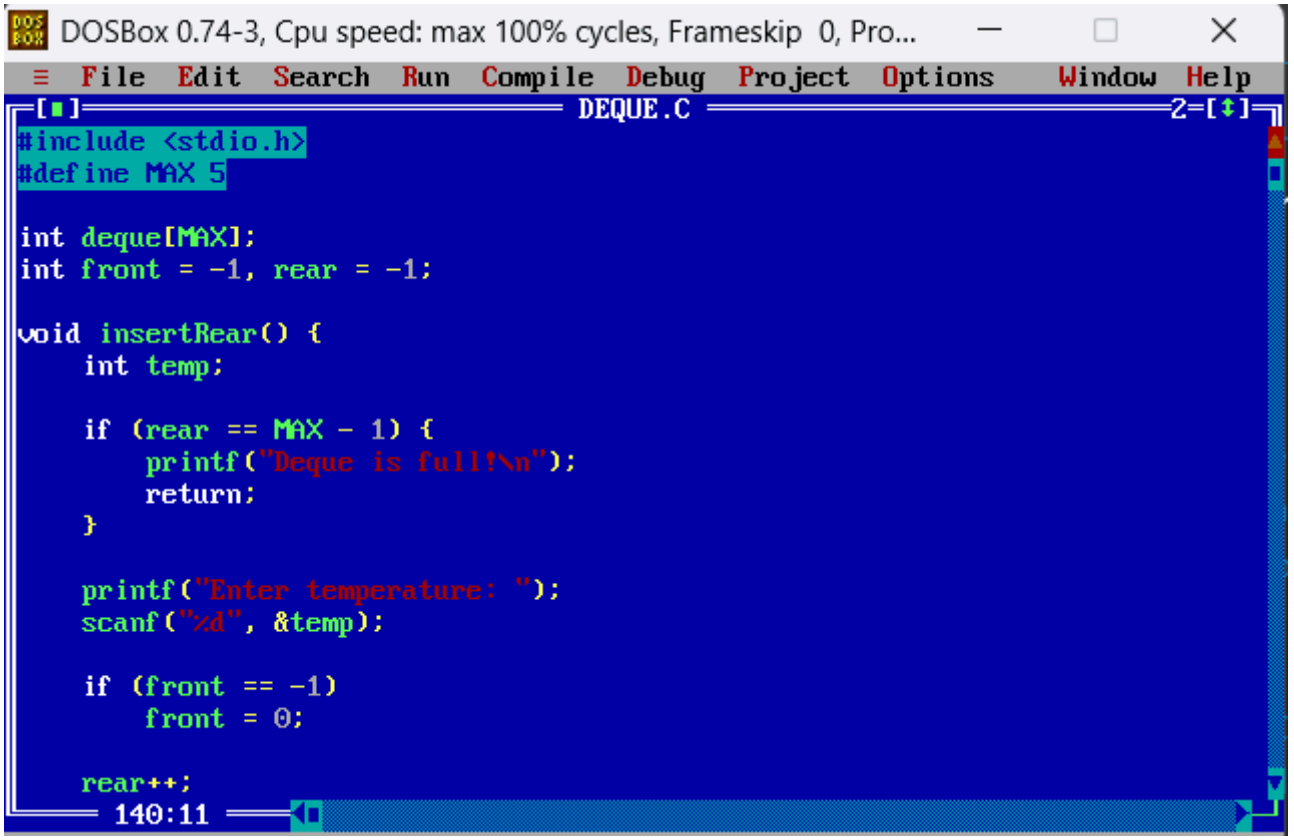
- Insert Front: Adds a temperature reading at the front.
- Insert Rear: Adds a temperature reading at the rear.
- Delete Front: Removes a temperature reading from the front.
- Delete Rear: Removes a temperature reading from the rear.
- Display: Displays all stored temperature readings.

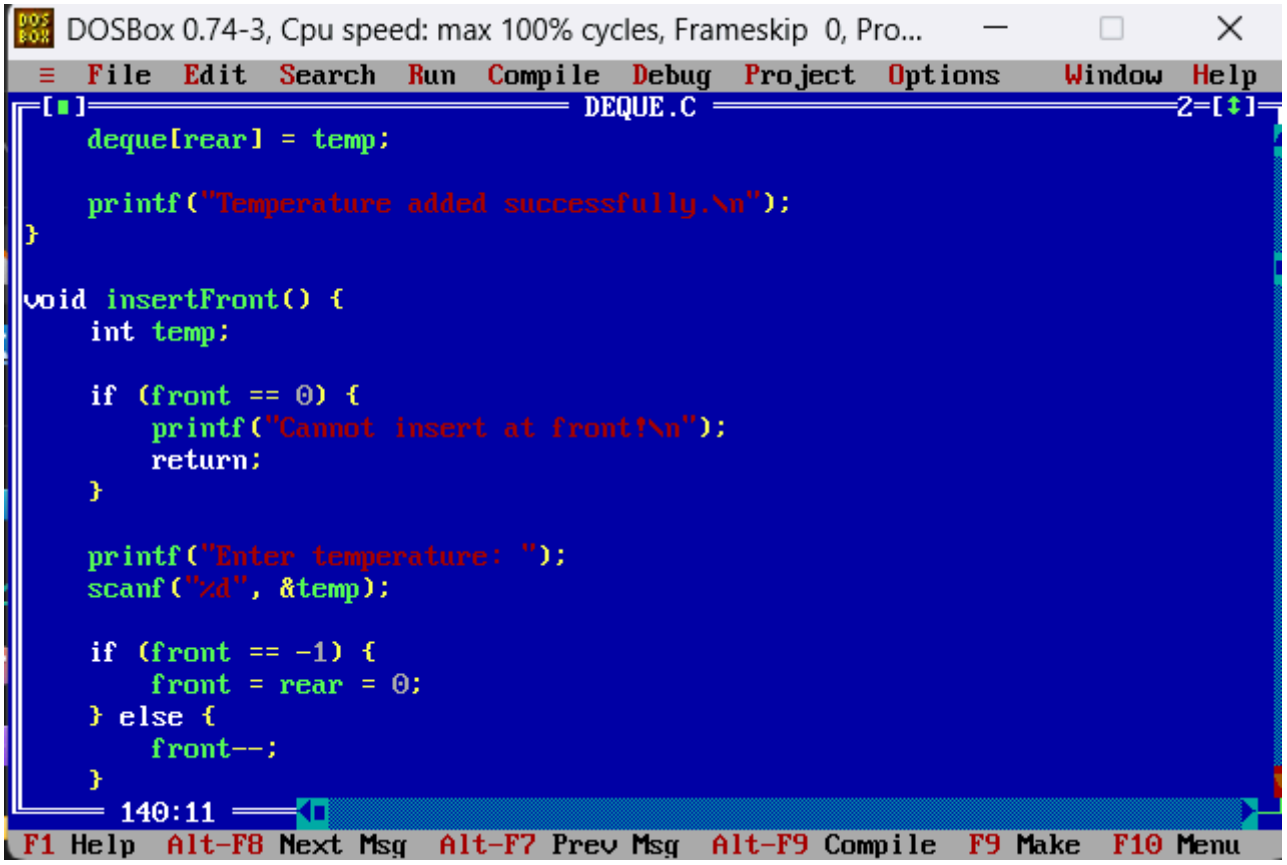
For example, if the readings are:

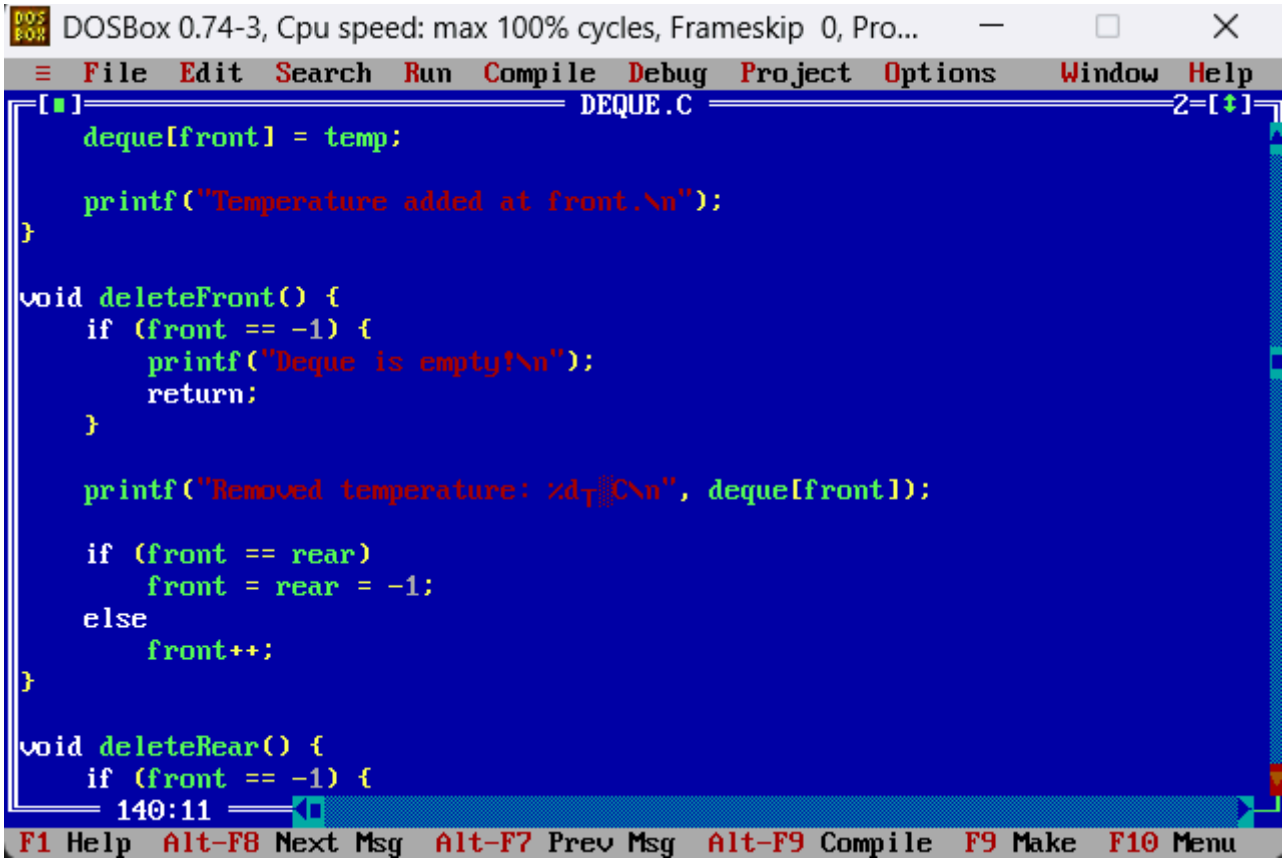
```
40°C → 42°C → 45°C
```

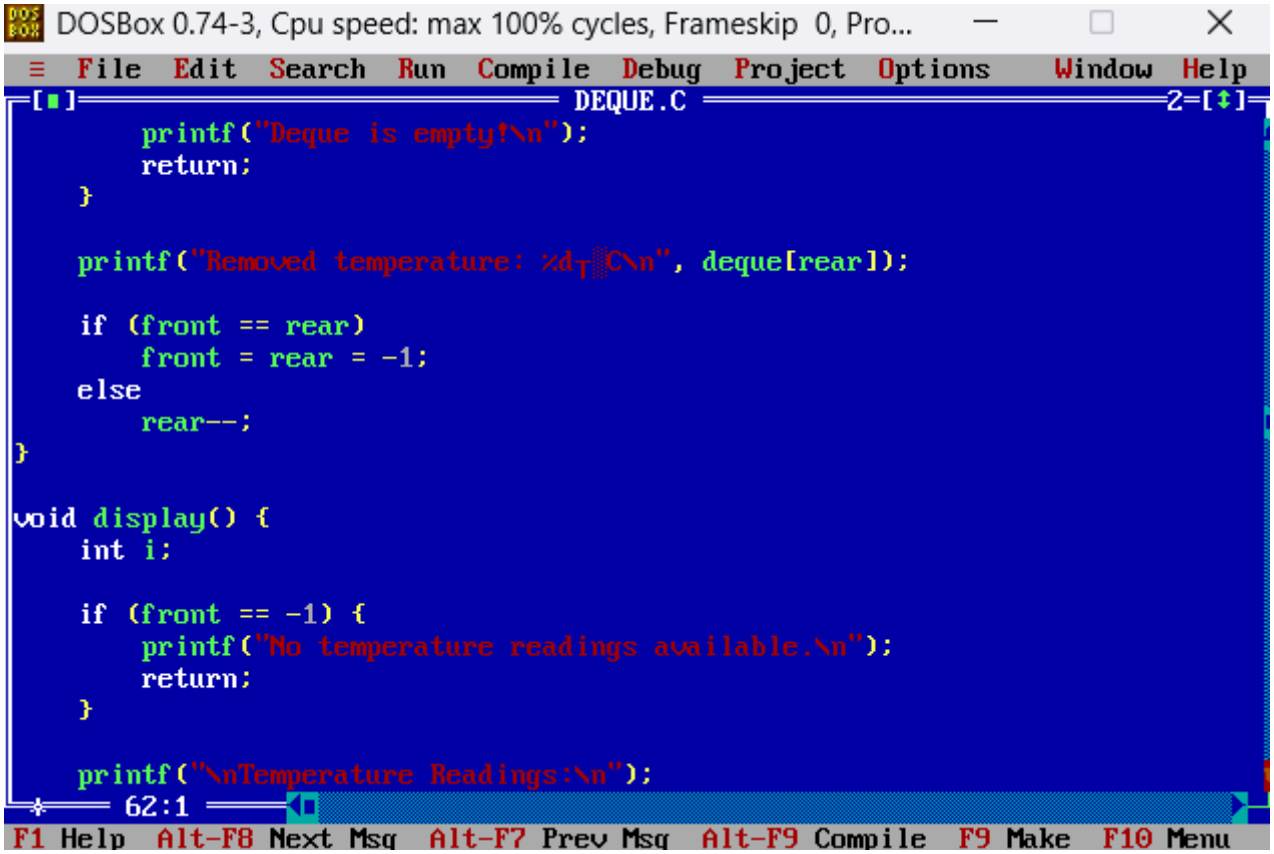
A new reading can be added at either the front or rear, and readings can also be removed from either end.

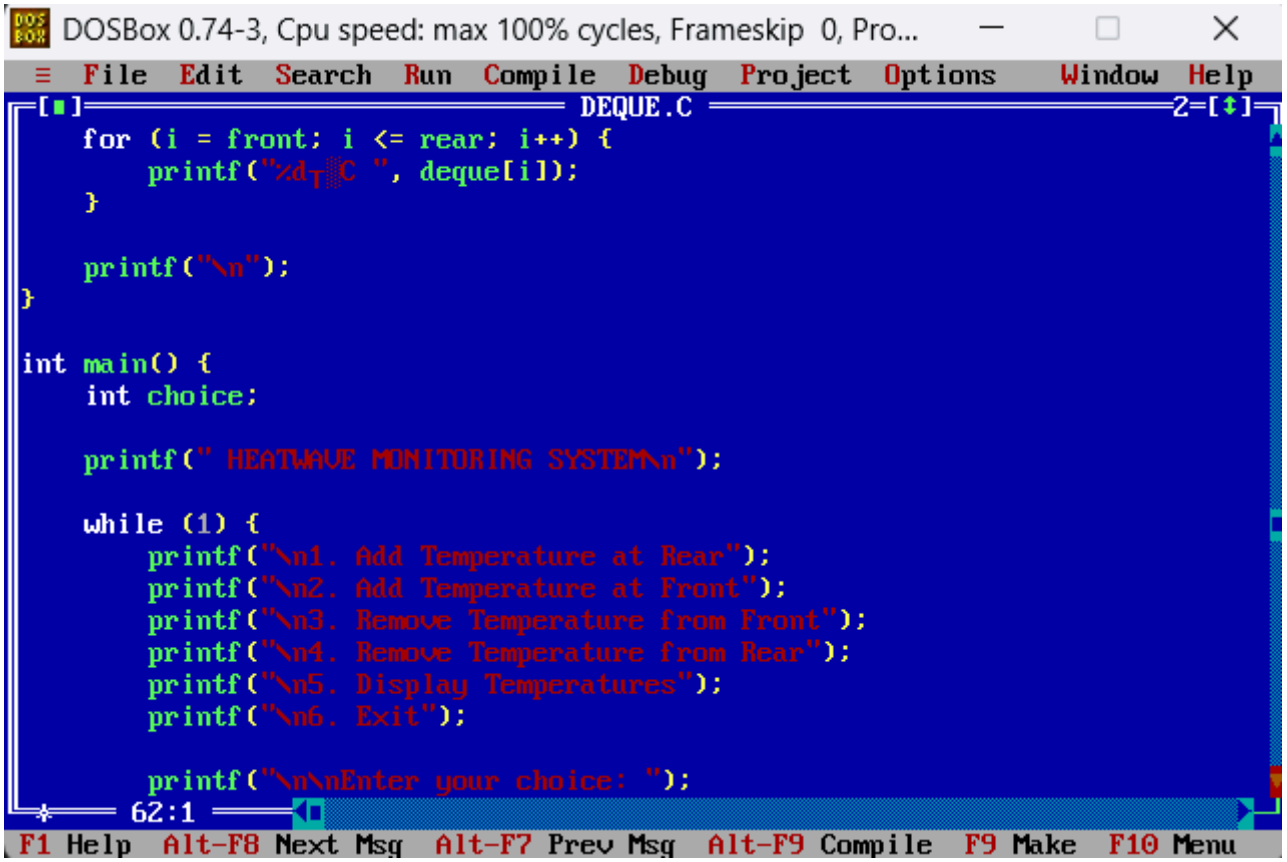
Code:

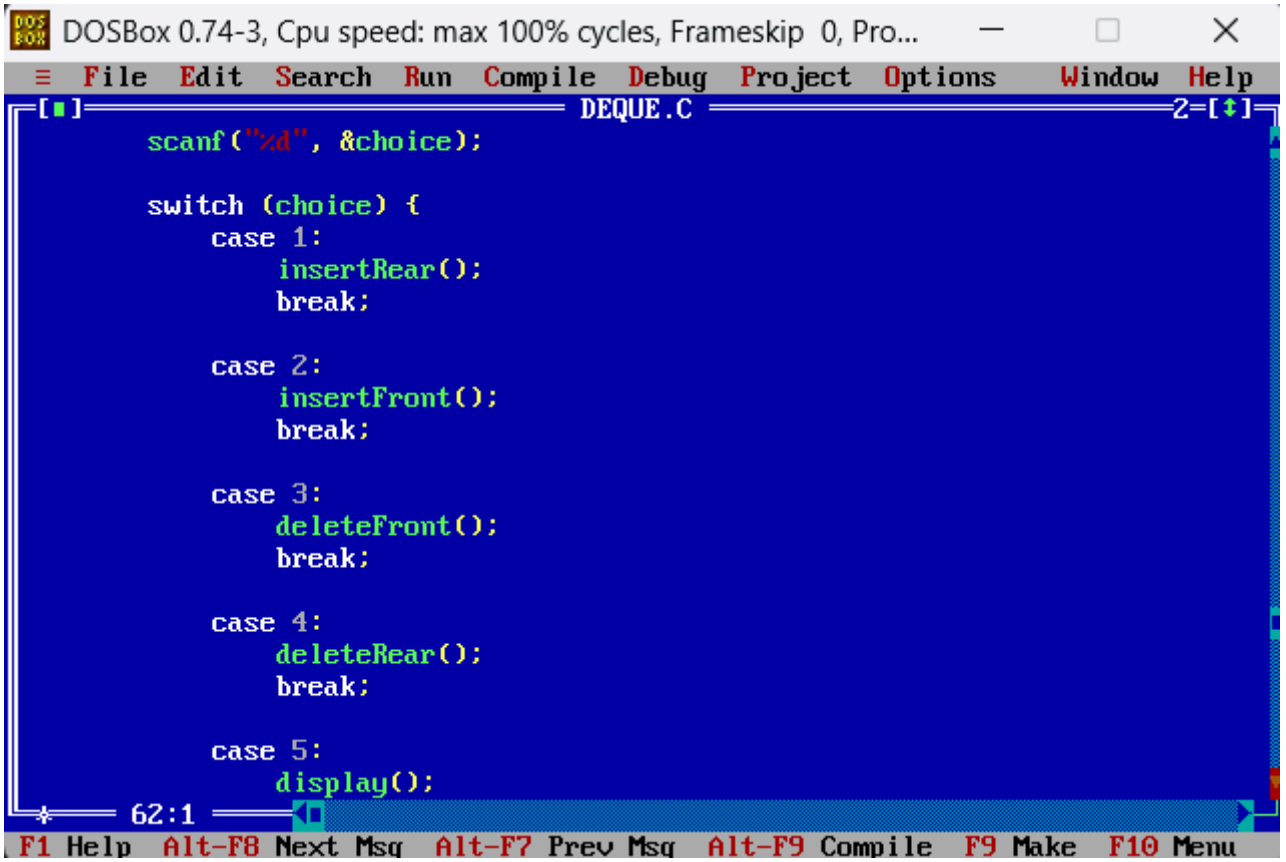

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
DEQUE.C
#include <stdio.h>
#define MAX 5
int deque[MAX];
int front = -1, rear = -1;
void insertRear() {
int temp;
if (rear == MAX - 1) {
printf("Deque is full!\n");
return;
}
printf("Enter temperature: ");
scanf("%d", &temp);
if (front == -1)
front = 0;
rear++;
140:11

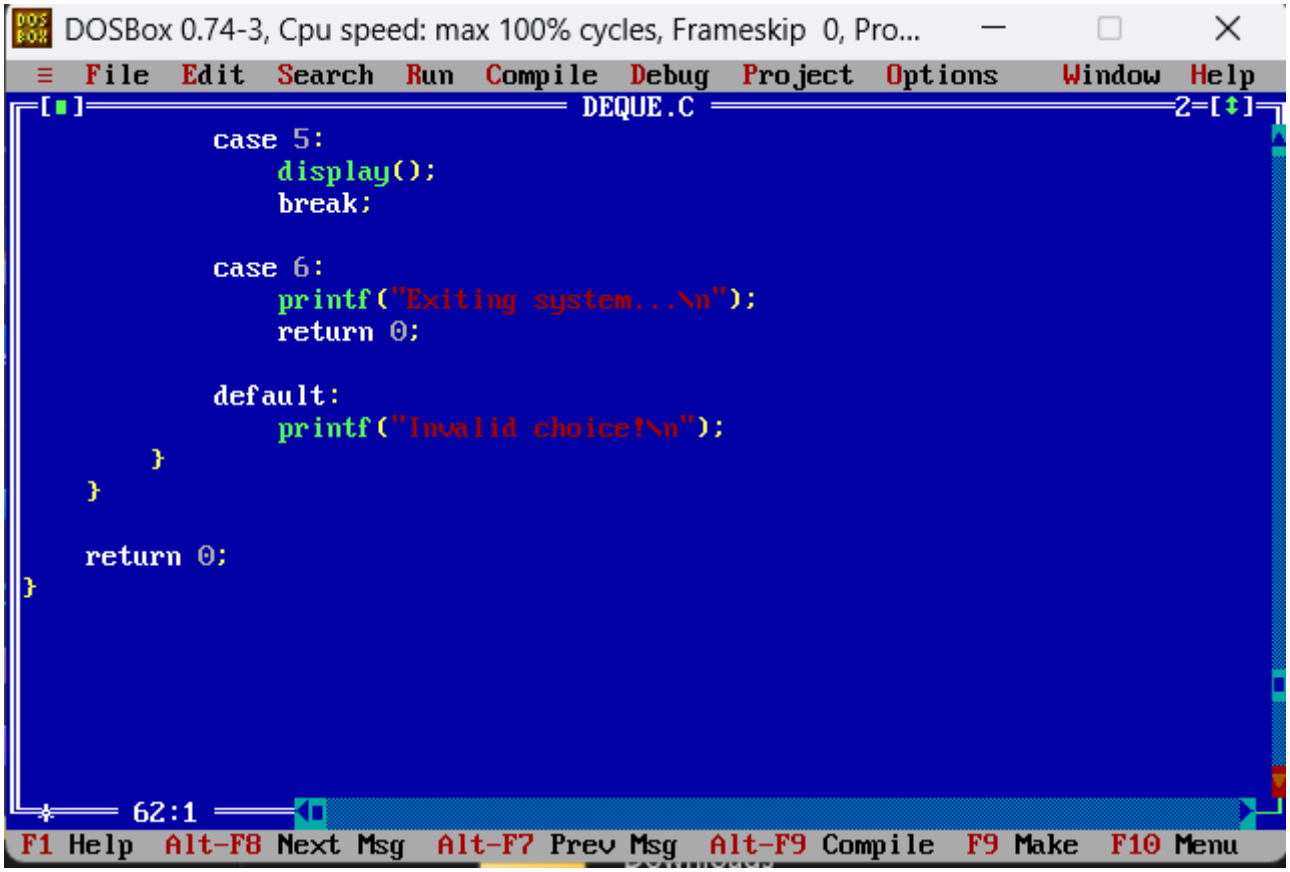

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
DEQUE.C
deque[rear] = temp;
printf("Temperature added successfully.\n");
}
void insertFront() {
int temp;
if (front == 0) {
printf("Cannot insert at front!\n");
return;
}
printf("Enter temperature: ");
scanf("%d", &temp);
if (front == -1) {
front = rear = 0;
} else {
front--;
}
140:11
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
DEQUE.C
deque[front] = temp;
printf("Temperature added at front.\n");
}
void deleteFront() {
if (front == -1) {
printf("Deque is empty!\n");
return;
}
printf("Removed temperature: %d°C\n", deque[front]);
if (front == rear)
front = rear = -1;
else
front++;
}
void deleteRear() {
if (front == -1) {
140:11
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

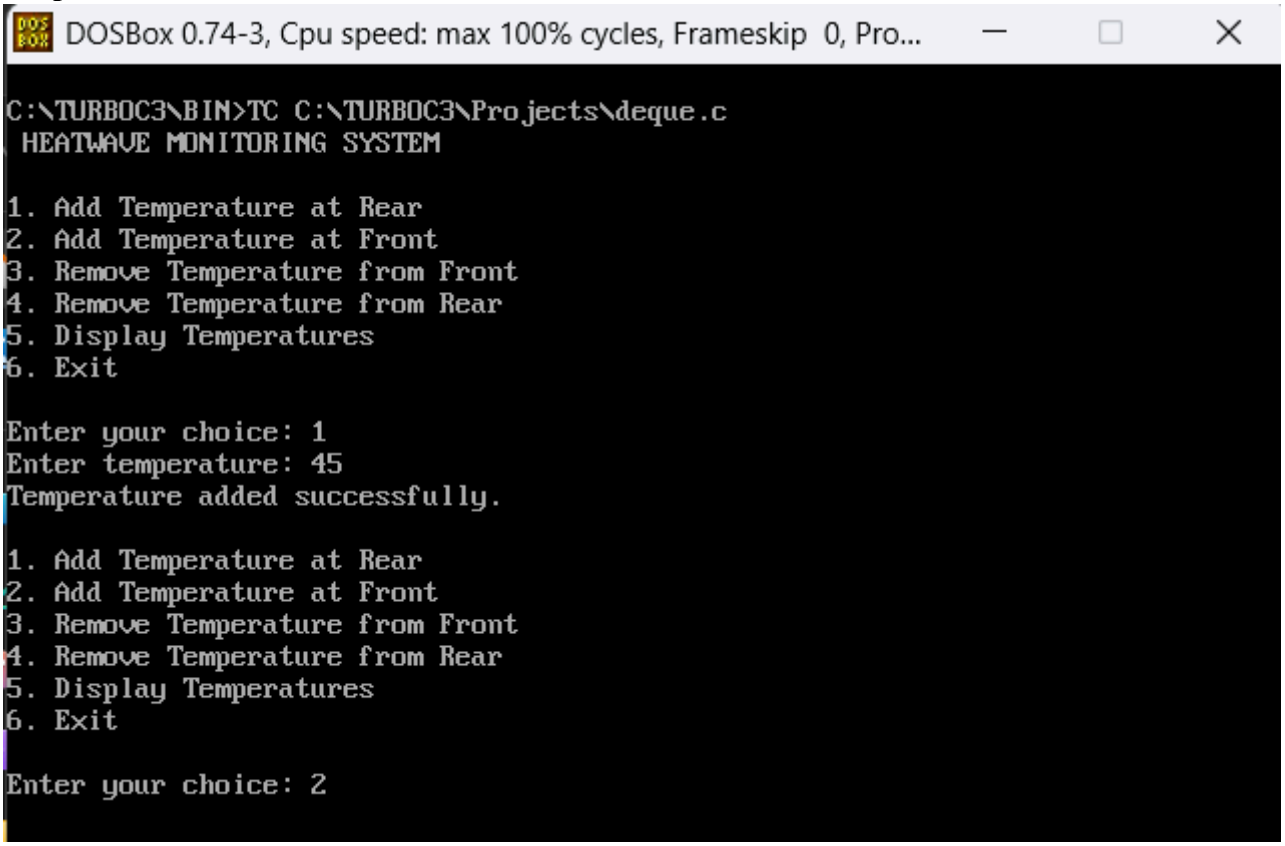

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
DEQUE.C
printf("Deque is empty!\n");
return;
}
printf("Removed temperature: %d°C\n", deque[rear]);
if (front == rear)
front = rear = -1;
else
rear--;
}
void display() {
int i;
if (front == -1) {
printf("No temperature readings available.\n");
return;
}
printf("\nTemperature Readings:\n");
62:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

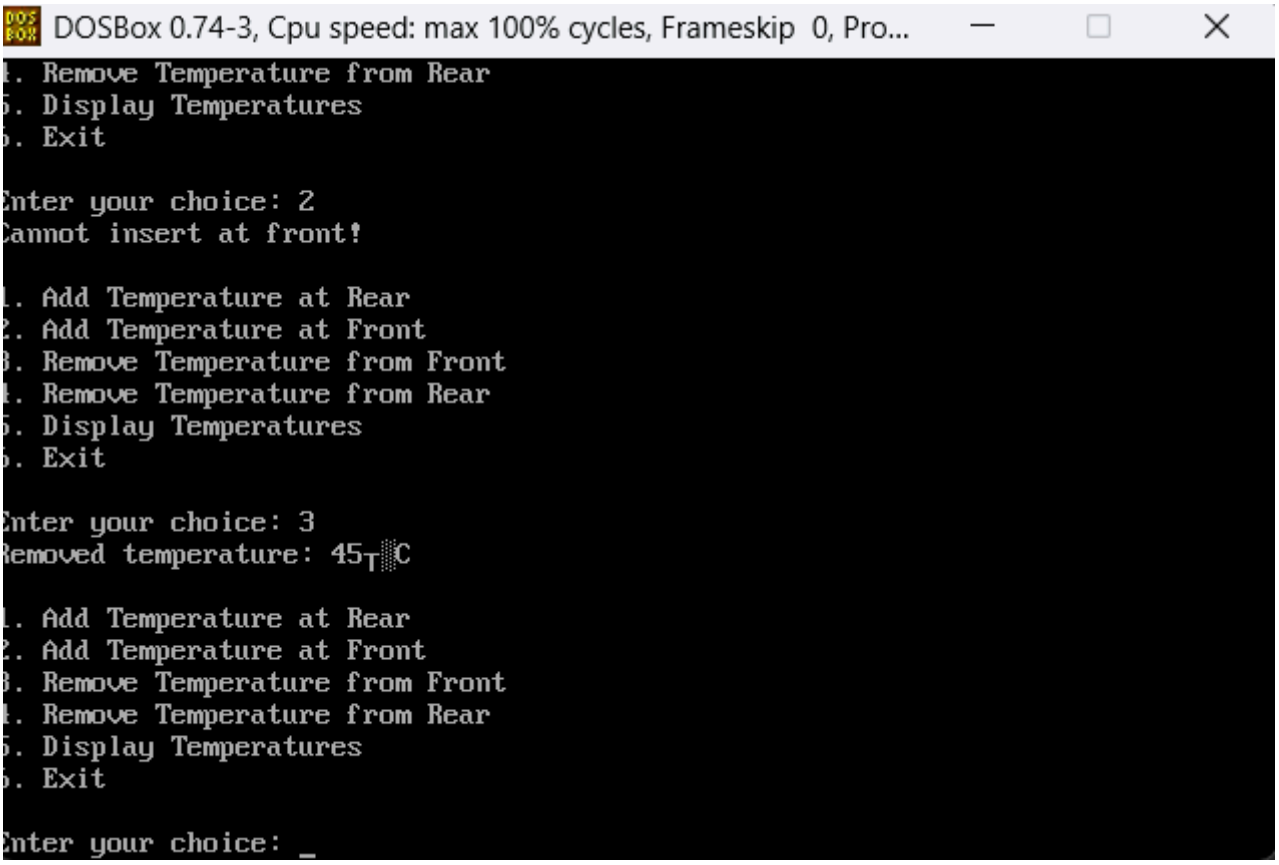

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
DEQUE.C
for (i = front; i <= rear; i++) {
printf("%d C ", deque[i]);
}
printf("\n");
}
int main() {
int choice;
printf(" HEATWAVE MONITORING SYSTEM\n");
while (1) {
printf("\n1. Add Temperature at Rear");
printf("\n2. Add Temperature at Front");
printf("\n3. Remove Temperature from Front");
printf("\n4. Remove Temperature from Rear");
printf("\n5. Display Temperatures");
printf("\n6. Exit");
printf("\n\nEnter your choice: ");
62:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

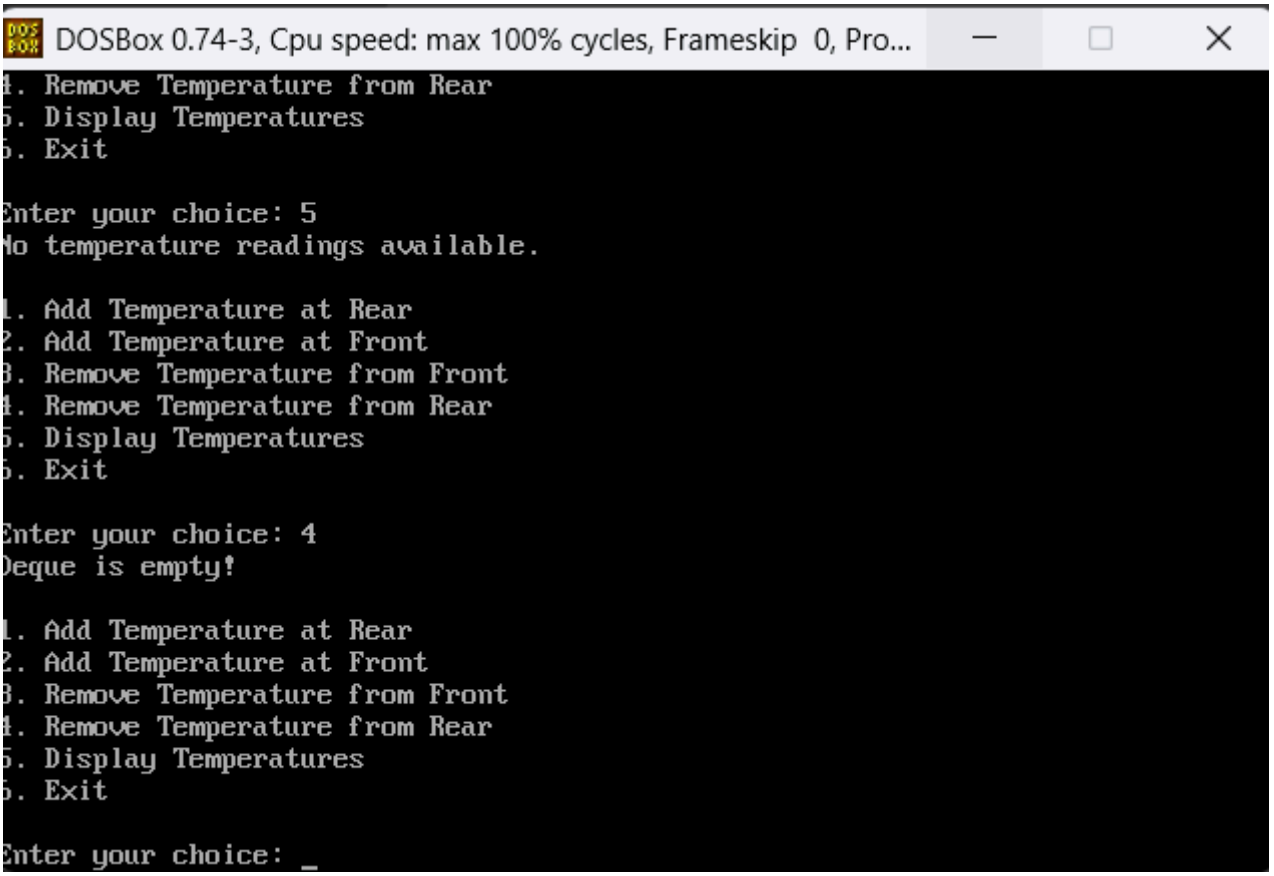

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
DEQUE.C
scanf("%d", &choice);
switch (choice) {
case 1:
insertRear();
break;
case 2:
insertFront();
break;
case 3:
deleteFront();
break;
case 4:
deleteRear();
break;
case 5:
display();
62:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
DEQUE.C
case 5:
display();
break;
case 6:
printf("Exiting system...\n");
return 0;
default:
printf("Invalid choice!\n");
}
}
return 0;
}
62:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

Output:


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
C:\TURBOC3\BIN>TC C:\TURBOC3\Projects\deque.c
HEATWAVE MONITORING SYSTEM
1. Add Temperature at Rear
2. Add Temperature at Front
3. Remove Temperature from Front
4. Remove Temperature from Rear
5. Display Temperatures
6. Exit
Enter your choice: 1
Enter temperature: 45
Temperature added successfully.
1. Add Temperature at Rear
2. Add Temperature at Front
3. Remove Temperature from Front
4. Remove Temperature from Rear
5. Display Temperatures
6. Exit
Enter your choice: 2


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
4. Remove Temperature from Rear
5. Display Temperatures
6. Exit
Enter your choice: 2
Cannot insert at front!
1. Add Temperature at Rear
2. Add Temperature at Front
3. Remove Temperature from Front
4. Remove Temperature from Rear
5. Display Temperatures
6. Exit
Enter your choice: 3
Removed temperature: 45
C
1. Add Temperature at Rear
2. Add Temperature at Front
3. Remove Temperature from Front
4. Remove Temperature from Rear
5. Display Temperatures
6. Exit
Enter your choice:



Conclusion: The Double-Ended Queue (Deque) was successfully implemented using an array for a Heatwave Monitoring System. The program allows the user to insert and delete temperature readings from both the front and rear ends. This experiment helped in understanding the working of Deque and its practical application in managing temperature data efficiently.

Outcome: After completing this experiment, the student will be able to:

- Understand and implement a Deque using an array.
- Perform insertion and deletion from both ends.
- Handle temperature data using user input.
- Apply the concept of Deque to a real-world Heatwave Monitoring System.
- Understand how arrays can be used to implement linear data structures.